

Audio-based Smart White Cane for Visually Impaired People

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Abstract—Despite the huge technological progress in digital control systems during the last decade, the problem of disabled people assimilating in modern society is still relevant due to the outdated technology in this area. One of the examples is the movement of visually impaired people. Nowadays, they use a white cane as a primary tool for interaction with the environment. The problem with this traditional technology is that the white cane does not give enough information about surrounding objects. Thus, the main modifications of the white cane would consist of integrating the obstacle detection system based on ultrasonic sensors, and informing the user by audio signals. The obstacle detection system is consist of a number of ultrasonic sensors located at different angles. Thus, they measure the distance from the cane to the ground. The data collected from the sensors, is analysed for three types of terrain: flat ground, small pit or hole, and stairs. In the case, when the sensors' reading does not match with the reading from the flat ground, the system will alert the user by the audio signal.

Keywords—Automation; control applications; visually impaired, smart device

I. INTRODUCTION

At present, more and more people are struggling with a visual disability. About 285 million people in the world are visually impaired [1]. Thirty-nine million of them are blind, and 246 million people have a low vision [4]. Among the many limitations faced by the blind human, the task of independent mobility and navigation is the most difficult. The mobility of the blind and visually impaired people is the combination of two related technologies: local navigation and global search for ways. Local navigation is the white cane, relying on ambient sounds, tactile cues from the surface of the ground and the use of guide dogs. Finding ways the questions of experience or training. Visually impaired people should rely on the help of others to find their way or require an accompanying person to be followed. Unfortunately, the number of blind people that are over 60 years old is growing by 2 million per decade. More worrying is that all these numbers are predicted to be doubled by 2020 [13]. Therefore, the development and improvement of assistive devices and systems for orientation and navigation is very important currently.

II. LITERATURE REVIEW

As it was aforementioned there is a huge issue of integration of visually impaired people into society; therefore, many pieces of research and projects were

conducted in order to create or modify the navigation system for easy integration of disabled people to society. There are different majority systems, which are focused on warning the person about obstacles. They are different in types of mounting to the person. The most popular are: sensors on hand, sensors on belts, sensors on helmet, and headsets [1]. However, the technology behind every work remains the same. It is possible to create the system to be attachable to the cane. First of all, the sensors should be selected. The most popular choices are infrared and ultrasonic sensors. For distance detection, the ultrasonic sensors were used [1–13] as it is low-cost. There are different implementations for this sensor. The majority of the researches used ultrasonic sensors as a detection of the obstacles in front of the person [1–4, 6–13]. In order to detect the side obstacles, additional sensors should be installed at the side of the cane. One of the implementations of ultrasonic sensors is detecting elevated obstacles, such as tables [5]. For this, ultrasonic sensors are located at the top part of the cane. It is possible to replace the ultrasonic sensors with infrared sensors [12]. The focus of the research with ultrasonic sensors is detecting the obstacles, but not defining the type of obstacles. Because of the low price, ultrasonic sensors would be used as the main obstacle detection for this work. As a microcontroller, there are two popular solutions. One is using Arduino, and second is creating their microcontroller, using ATmega328 chip. Every method has its advantages. Using Arduino reduces time spent on the work with microcontroller [1, 6–9, 11–12]. Arduino is shipped with its software, which is using object-oriented language C++. The compiler has a user-friendly interface and has big functions library, which drastically reduces time, spent on the programming [1, 6–9, 11–12]. The advantages of creating their microcontrollers are low price and flexibility in functions [2–5]. The flexibility of this method allows creating own schematics for specific purposes, which would be more efficient than Arduino. However, our research is not so heavy influenced by the efficiency of microcontrollers. Moreover, the time, which we could spend on this research, was limited. Therefore, an Arduino was used as the microcontroller for this research. As an actuator, there are two popular choices. The first one is the speaker. It is used in order to tell the person about incoming obstacles. In order to recreate simple sounds, the simple setup is enough. Any speaker could be connected to Arduino directly [2, 3, 5, 6, 11]. The internal memory is enough in order to play a simple sound. However, the goal of our work is to play the voice. Thus, the SD card was used in order to record the sounds, store it, and play when it is required [11]. The

second type of actuators is the DC motor. According to the reference papers, the DC motor is connected directly to Arduino. The rotation of the motor generates vibrations. The intensity of the motor is depended from a distance between cane and obstacles [1, 4, 7, 9]. The motor could not represent different types of obstacles. Thus, the speaker with SD card was used for this research. All mentioned above information is common for almost all reference papers. However, there are unique modifications in the navigation systems, such as water sensors, GPS, a monocular camera, light sensor, humidity sensor, and temperature sensor. The water sensor is the most popular modification. It is fairly simple to implement. The sensor consists of two connectors. When the cane is touching a water pound, the current will go through the connectors, signifying water [2, 18–20, 23]. The second most popular modification among the reference papers is GPS navigation. The primary usage of GPS navigation is to navigate the person in the city [1, 9]. This navigation system commonly uses voice notification when it is time to change the direction. However, it is possible to create a plastic implant to the mouth of the person [1]. This implant has connectors, located in the circle. When it is required to change the direction, the implant sends little current in the required direction. The last modification is light, temperature, and humidity sensors. The main purpose is to help in home life. The light sensors are telling whether the light in the room is on or off. The last two sensors are served to give general information to the person, like temperature and humidity. In conclusion of the literature review, the majority of the sources are using ultrasonic sensors as a primary navigation sensor. The actuator is dynamic. The focus of the reference papers is whether in finding obstacles in front of the cane or finding elevated obstacles. None of them is focused on combining this technology in one cane. The modification is possible; however, it requires additional time.

III. PROTOTYPE DEVELOPMENT

This work consisted of three main steps: purchasing main components, different programming systems, and assemble the systems into one equipment. First of all, the consumables were obtained. The components are divided into two categories: primary and secondary. With only the primary components, the work could be started. Arduino, wires, and ultrasonic sensors are included in the primary category. For the secondary category, all the rest of the sensors and actuators are included. After that, the second step had been started.

It should be noted, that as soon as primary components are delivered, both step one and step two could be done in parallel. In other words, the work could be done using primary components at the same time, as the secondary components are not obtained. Using primary components, the obstacle detection system was programmed. The only sensor type in this system is the ultrasonic sensor. The output of this sensor is analog. It means that the sensor outputs the difference in voltage. Thus, the system should be created in order to decode the voltage change into three types of output: no obstacle, high obstacle (stairs, wall), low obstacle (hole in the ground). The noise was eliminated using pre-filter;

therefore, the pre-filter was designed. The next step was the application of PID controller, in order to have a correct signal. The final step was coding the behaviour of the microcontroller depending on the sensor output. The microcontroller will be searching in the detection of a sudden change in sensor output. Depending on whether the signal output is sudden decreased or increased, the obstacle would be high or low. If there is no or small change of signal (when the person changes the angle of the cane), the obstacle type would be none. There is the final system, which was implemented.

The user can get a voice warning using a headset. The voice warning requires an SDcard reader, SD card, and headset. The SD card is required for storing the pre-recorded warning audio files. The SD card reader is required for reading data from SD card. The code for Arduino was created, which plays audio files from the headset. Moreover, the code chooses audio files depending on the output of the ultrasonic sensor. The final step was assembling every system in one. The systems are mounted on the cane. Any stick of the similar could play the role of the cane to the cane length. The standard stick was chosen.

A. Components Selection

The main components used in developing the audio-based white cane prototype is listed from *a* to *e*. The components are cheap and easily available. Thus; the prototype development cost is low.

- a) Ultrasonic sensor
- b) Arduino Uno
- c) DC motor
- d) SD card reader
- e) Headset

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is no or small change of signal (when the person changes the angle of the cane), the obstacle type would be none.

For the next step, the availability of the secondary components is required. There are two more systems, which should be implemented. The first one is voice warning using a speaker, and the second one is colour detection. The voice warning requires an SD card reader, SD card, and speaker. The SD card is required for storing the pre – recorded warning audio files. The SD card reader is required for reading data from SD card. The code for Arduino should be created, which will play audio files from the speaker. Moreover, the code should choose audio files depending on the output of the ultrasonic sensor.

B. Conceptual Design

The last system is colour detection. There is only one sensor, which consists of 4 white LED and lighted sensible sensor. The code should convert analogue output to output colour type. It would be done through experiments. For example, the output of the sensor would be analysed for different tones of red. Thus, we will find at what output range the red colour is. The same procedure will be continued for every colour. The final step is to assemble every system in one. The systems would be mounted on a cane. Any stick of the similar could play the role of the cane to the cane length. The obstacle detection system should be recreated three times for the right, front, and left sides. The colour sensor and the speaker could be mounted near the handle.

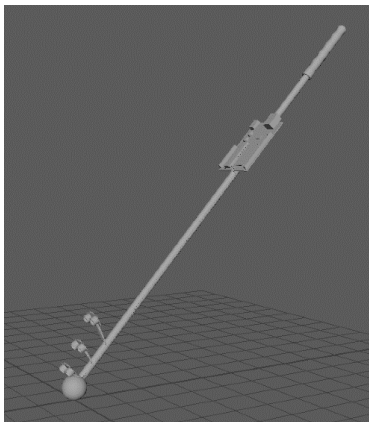


Figure 1. Isometric view of the prototype

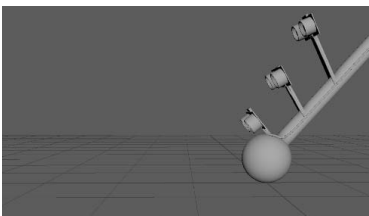


Figure 2. Enlarge view of the bottom part

IV. RESULTS AND DISCUSSIONS

After completing all the steps to design the following obstacle detection system was obtained as highlighted in Fig. 3.



Figure 3. Front view of the prototype

Two different voice messages such as “high ground obstacle” and “low ground obstacle” are used to alert the user. Testing results of these examples can be observed from Figure 4. The system detects low ground obstacle if the difference between two corresponding distance measures lower than -50 mm. In the following case (see Figure 10), the difference between N1 (119) and N2 (170) is -51.

The high ground obstacle is obtained when the difference between points is higher than 50 mm. In the following case, the difference between N1 (173) and N2 (116) is 57. During the testing, the proposed prototype successfully addressed all voice alerts. However, it is the prototype model, and an advanced model will have broader obstacle range.

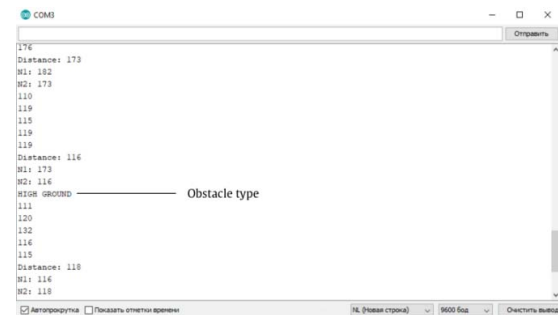


Figure 4. High ground obstacle

The noteworthy thing is that arithmetical mean of the last five distance measurements is taken in order to provide very precise distance measurement (see Figure 5).

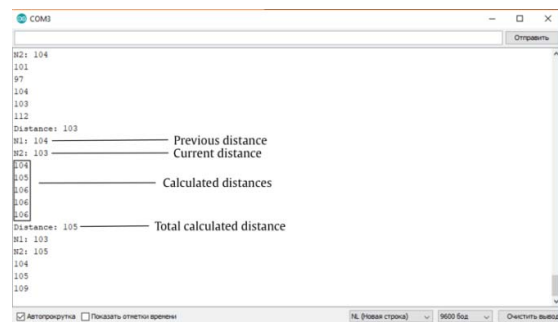


Figure 5. Distance calculation

A. Algorithm Development

The code consists of 4 main parts: variables declaration, setup function, loop function, and set of other functions. In the first part, all libraries and variables are declared. The majority of variables are integers for pin information storage or boolean variables for checking conditions. In the second part, the setup function should be written. This function starts once at the moment of turning on the Arduino. Thus, at this function, all pins are declared using variables which were declared in the first section. In the third section, the loop function should be coded. This is the core function, because it loops all the time, while the Arduino is turned on. In this section, all custom functions are called. In this case, the loop function calls only 3 functions: Button1(), Button2(), and obstSelect(). The Button1() and Button2() functions are called in order to check whether buttons are pressed or not. These functions were written after the loop function. The third function is obstSelect(). This function is called only when the sensor is in a horizontal position. In order to check if the sensor is in a horizontal position, boolean "hor" is checked. This boolean is changing by functions Button1() and Button2(). The last part of the code is the longest. This part contains all custom functions which are called by the loop function. The obstSelect() function is used in order to define whether the obstacle is high ground (step) or low ground (hole). This function takes the value from the other function called DistanceCount(). This function outputs the distance, measured by an ultrasonic sensor. The distance is stored at variable and then compared with the previously measured distance. If the distance is less than 5 cm, then no obstacle is defined. When the difference is higher than 5 cm, the Arduino sends audio signal through the headsets by informing the high ground or low ground, depending from difference sign. The output distance from the ultrasonic sensor is varying by too much. Thus, in order to minimise errors, the mean value was taken. The role of the DistanceCount() function is to count the distance five times and take mean value. This was done with for loop. The DistanceCount() function measures the function ServoMot().servomotor() function measures the time, taken for the signal to travel from the ultrasonic sensor and back. Knowing the speed of sound, we wrote the formula for calculating distance in mm. The Button1() and Button2() functions are used in order to start and end the obstacle search process. When the button is pressed, the signal travels to the Arduino, which makes the required Boolean to change the value from "false" to "true". The last 3 custom functions are obLow(), obHigh(), and Hello(). These functions have the only purpose of playing required sound. These functions are called in order to inform the person about obstacle type and to turn the system on.

V. PERFORMANCE AND LIMITATIONS OF THE PROTOTYPE

The cane has an ON-OFF switch to save the batteries. There are a few insignificant limitations for the developed prototype. The prototype can run for half an hour with a 9v DC battery. We can add a 12-volt battery to the prototype, but the overall weight will increase significantly. Two 9V

rechargeable batteries can help the cane to run for an hour. The dimension of the cane is 1.2 m long, and the diameter of the cane is 30 mm. The code was uploaded into Arduino using official software. Initially, only distance measurement capabilities of the ultrasonic sensor were tested. According to the original design, the ultrasonic sensor should be located on some angle from the floor. However, the sound wave was reflected from the surface, and the sensor would show the wrong distance (Figs. 6 and 7). In order to confirm this theory, we tried to block possible reflected sound wave, and the distance changed. Thus, the sound wave was reflected from the floor, and therefore, the new distance measurement techniques should be implemented.

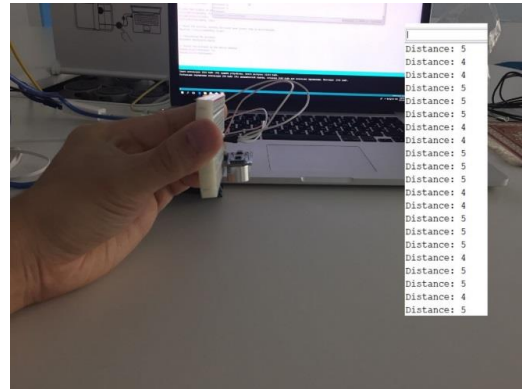


Figure 6. The perpendicular position of the sensor

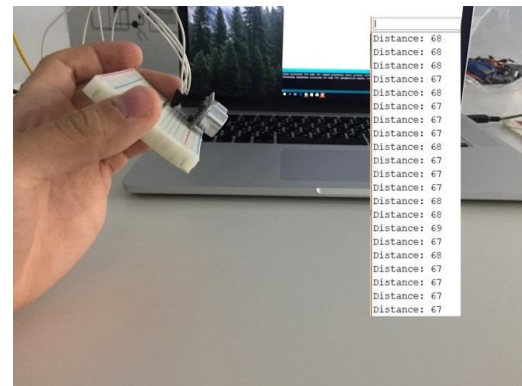


Figure 7. Angled position of the sensor

According to the new design, the sensor will be located perpendicular to the ground. The sensor will be located on the stick. Initially, the stick will be located on the cane. As soon as a button is pressed, the servo motor will rotate the stick with the sensor on the end until the sensor is in the horizontal position. For this test, the sensor was rotated manually, and the behaviour of the motor was observed. As soon as the sensor was in a horizontal position, the motor stopped working. Thus, the new code works correctly.

VI. CONCLUSIONS AND FUTURE PERSPECTIVE OF THE PRESENT WORK

The fundamental aim of the design was to develop the obstacle detection system for visually impaired people with

the aid of ultrasound sensors and by the white cane. According to tests carried out on its functions and working conditions reveal that a proposed model achieved objectives planned by the team. This detection system mounted on the stick is a practically viable product and appropriate to carry around like other walking sticks. The designed model can be more equipped with other sensors to make it a flawless device for the visually impaired people. For example, water or colour sensors could be implemented into the system to enhance the functionality of the proposed model. Future work will focus on refining the performance of the prototype. Another outstanding improvement could be a modification of the system based on the feedback of visually impaired people. They can point out shortcomings and highlight the best aspects of the proposed system.

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